HVAC DESIGN CALCULATION REPORT

Sample Villa - Riyadh

Riyadh, Saudi Arabia

Building type: villa - 2 floor(s)

Code basis: Saudi Building Code + ASHRAE | Units: SI

Issue: P0 (preliminary) 2026-10-15

PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

1. Design basis

1.1 Codes and standards

Ref	Code / standard
SBC501	SBC 501 Saudi Mechanical Code (2018, as amended)
SBC601	SBC 601 Saudi Energy Conservation Code - Commercial
SBC602	SBC 602 Saudi Energy Conservation Code - Residential
A62.1	ANSI/ASHRAE Standard 62.1-2019 Ventilation for Acceptable IAQ
A90.1	ANSI/ASHRAE/IES Standard 90.1-2019 Energy Standard
AF	ASHRAE Handbook - Fundamentals (CLTD/SCL/CLF method 1997 Ch.28; climate data 2021 Ch.14)
SMACNA	SMACNA HVAC Duct Construction Standards - Metal and Flexible, 3rd ed. 2005

1.2 Design conditions

Parameter	Value	Basis
Outdoor summer design (0.4 %)	45.6 °C DB / 19.4 °C MCWB, W = 4.4 g/kg	ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK
Daily range	14.6 K	ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK
Outdoor winter design (99.6 %)	6.6 °C	ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK
Indoor cooling	24.0 °C, 50 % RH (W = 10.0 g/kg)	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)
Indoor heating	21.0 °C	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)
Site elevation / pressure	612 m / 94.19 kPa; air factors 1.143 W/(L/s·K), 2798 W/(L/s·kg/kg)	ASHRAE Fundamentals Ch.1
Latitude / design month	24.71° / July (CLTD tables)	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36

Code reference: SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)

1.3 Building envelope

Element	U-value / property	Basis
External wall	0.40 W/m²K (CLTD wall group D)	SBC602
Roof	0.20 W/m²K (roof_insulated_concrete)	SBC602
Glazing	2.67 W/m²K, SHGC 0.25 (SC 0.29)	SBC602
Slab edge (heating)	F = 1.2 W/m·K	SBC602

Code reference: SBC 602 Ch.4 prescriptive envelope table / SBC 601 Ch.5 (max U-factor, SHGC)

1.4 Internal loads

Space type	Occupancy	Activity	Lighting W/m²	Equipment W/m² + fixed	OA category	Exhaust	Cooled
bath	0 persons	seated	8	0 + 0 W		toilet_private	no
bedroom	2 persons	sleeping	8	5 + 0 W	dwelling		yes
corridor	0 persons	seated	6	0 + 0 W	corridor		yes
dining	2.5 m²/person	seated	10	3 + 0 W	dwelling		yes

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dressing	0 persons	seated	8	0 + 0 W	dwelling		yes
hall	1 persons	seated	8	0 + 0 W	corridor		yes
kitchen	2 persons	light_work	10	0 + 1500 W	dwelling	kitchen_residential	yes
laundry	0 persons	light_work	8	0 + 0 W		laundry	no
living	4.0 m²/person	seated	10	8 + 0 W	dwelling		yes
maid_room	1 persons	sleeping	8	5 + 0 W	dwelling		yes
majlis	2.5 m²/person	seated	12	4 + 0 W	dwelling		yes
master_bedroom	2 persons	sleeping	8	6 + 0 W	dwelling		yes
shaft	0 persons	seated	0	0 + 0 W			no
stair	0 persons	seated	6	0 + 0 W	corridor		yes
store	0 persons	seated	5	0 + 0 W			no

Code reference: ASHRAE Fundamentals 2021 Ch.18 Table 1 (people), Tables 2-3 (lighting), Tables 5-9 (equipment)

1.5 Assumptions

All assumptions are logged in assumptions.md. Items marked 'confirm' must be confirmed by the engineer of record.

ID	Step	Assumption	Source	Status
A001	A-read	Page 1 scale 1:100 (title+dimensions, confidence high)	scale_detect.py	default
A002	A-read	Page 1 north = 19.6 deg clockwise from page-up (detected from north arrow)	pdf_reader.py	default
A003	A-read	Page 2 scale 1:100 (title+dimensions, confidence high)	scale_detect.py	default
A004	A-read	Page 2 north = 19.6 deg clockwise from page-up (detected from north arrow)	pdf_reader.py	default
A005	A-read	Page 3 scale 1:100 (title+dimensions, confidence high)	scale_detect.py	default
A006	A-read	Page 3 north = 19.6 deg clockwise from page-up (detected from north arrow)	pdf_reader.py	default
A007	B-loads	Outdoor design 45.6 °C DB / 19.4 °C MCWB (0.4 %), daily range 14.6 K, heating 6.6 °C (99.6 %), elevation 612.0 m (ASHRAE 2021 Ch.14 (0.4% cool / 99.6% heat) - OERK, approximate - verify)	references/climate_data.csv	default
A008	B-loads	Indoor design 24.0 °C / 50 % RH cooling, 21.0 °C heating	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)	default
A009	B-loads	Envelope U-values = code maximum (SBC602: wall 0.4, roof 0.2, glazing U 2.67 / SHGC 0.25 W/m²K) - actual build-up not given	SBC 602 Ch.4 prescriptive envelope table / SBC 601 Ch.5 (max U-factor, SHGC)	CONFIRM
A010	B-loads	Cooling loads by CLTD/SCL/CLF; CLTD tables for wall group D, roof 'roof_insulated_concrete', latitude row nearest 24.71°, July; colour factor K=0.83 (medium walls), roof K=1.0; glass with interior shading=True	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36	CONFIRM
A011	B-loads	CLTD/LM/SHGF/CLF tables in references/cltd_tables.json are transcribed-approximate - verify against licensed ASHRAE 1997 Ch.28 tables	references/cltd_tables.json	CONFIRM
A012	B-loads	CLF for people/lights/equipment = 1.0	ASHRAE Fundamentals 1997 Ch.28	default

		(24 h operation, no storage credit)	CLTD/SCL/CLF method, Tables 30-36	
A013	B-loads	Partitions/floors between conditioned and exhaust-only rooms (baths, stores) not counted: those rooms are near room temperature	designer judgement	default
A014	B-loads	Outdoor humidity ratio 4.4 g/kg < indoor 10.0 g/kg at design: outdoor-air latent load taken as 0 (conservative); check a dehumidification design day	ASHRAE Fundamentals 2021 Ch.14 (DP design)	CONFIRM
A015	B-loads	Supply air 13.0 °C (room-supply ΔT 11.0 K) for airflow	designer default	default
A016	B-loads	Infiltration by air-change method: {'one_exterior_side': 0.5, 'two_plus_exterior_sides': 0.75, 'no_exterior': 0.0} ACH (residential)	ASHRAE Fundamentals 2021 Ch.16 §'Residential air change method'	default
A017	B-loads	Occupancy, lighting and equipment densities from references/space_types.json	ASHRAE Fundamentals 2021 Ch.18 Table 1 (people), Tables 2-3 (lighting), Tables 5-9 (equipment)	default
A018	B-zoning	Zoning: own zone for ['majlis', 'master_bedroom']; other ducted rooms grouped by position per floor up to 17.6 kW per unit (incl. safety factor)	references/design_defaults.json zoning	CONFIRM
A019	B-zoning	Rooms exhaust-only (not cooled; make-up by transfer air): GF-09, GF-11, GF-13, FF-01, FF-02, FF-12, FF-14	references/space_types.json	CONFIRM
A020	B-vent	Z-GF-01 (KITCHEN) has no exterior wall: its 14 L/s OA is supplied via Z-GF-04 and transferred	SBC 501 §403.3.1.1 & Table 403.3.1.1; ASHRAE 62.1-2019 §6.2.2.1 Eq.6-1, Table 6-1	CONFIRM
A021	B-vent	Zone air distribution effectiveness Ez = 0.8 (heat-pump heating with ceiling supply of warm air)	ASHRAE 62.1-2019 §6.2.2.2 Table 6-2 (Ez)	default
A022	B-vent	Exhaust fans sized for the intermittent (rated) rate; continuous operation would allow the lower rate	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)	default
A023	B-equip	Safety factor 10% on cooling and 10% on heating, applied to coil loads at selection (shown in tables)	Designer allowance - not a code requirement	CONFIRM
A024	B-equip	Nominal (T1) capacities derated x0.88 for 45.6 °C outdoor design (T3) - replace with brand T3 data	references/equipment_catalog.json t3_derate	CONFIRM
A025	B-equip	Equipment from GENERIC catalogue (references/equipment_catalog.json) - replace with the approved brand	references/equipment_catalog.json	CONFIRM
A026	B-equip	Minimum supply per served room 30 L/s (smallest diffuser)	references/air_terminals.json	default
A027	B-equip	GF-06 KITCHEN has no exterior wall: wall split on interior wall, refrigerant/condensate run in ceiling	designer judgement	CONFIRM
A028	B-equip	IU-GF-02 (WS-09) is 141 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A029	B-equip	Z-GF-03 indoor unit in the ceiling void of GF-02 ENTRANCE HALL (service space next to the zone)	references/design_defaults.json unit_host_preference	CONFIRM
A030	B-equip	Z-GF-04: sensible capacity governs (8.2 kW sensible at catalogue SHR) - DS-48 chosen instead of DS-36; brand data at actual (dry) entering conditions may allow the smaller unit	designer note	CONFIRM
A031	B-equip	Z-GF-04 indoor unit in the ceiling void of GF-04 CORRIDOR (service space next to the zone)	references/design_defaults.json unit_host_preference	CONFIRM

A032	B-equip	IU-FF-01 (DS-24) is 127 % above requirement - smallest catalogue size; consider combining zones or a smaller (VRF/wall) unit	designer note	CONFIRM
A033	B-equip	Z-FF-01 indoor unit in the ceiling void of FF-04 DRESSING (service space next to the zone)	references/design_defaults.json unit_host_preference	CONFIRM
A034	B-equip	Z-FF-02 indoor unit in the ceiling void of FF-07 LOBBY (service space next to the zone)	references/design_defaults.json unit_host_preference	CONFIRM
A035	B-equip	Outdoor units grouped on the roof near the riser/shaft with 0.6 m service clearance; check structure, parapet height and acoustic screening	designer judgement	CONFIRM
A036	B-ducts	Ducts sized by equal friction 0.8 Pa/m with velocity limits {'main': 5.0, 'branch': 3.5, 'runout': 2.5, 'return': 3.5, 'fresh_air': 4.0, 'exhaust': 4.0} m/s; max duct depth 300 mm from ceiling void	ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal)	default
A037	B-ducts	Fittings allowed as +50% equivalent length; unit plenum/filter 40.0 Pa	designer default	CONFIRM
A038	B-ducts	Duct routes are schematic trunk-and-branch lines over the plan; coordinate with structure (beams), lighting and other services	designer judgement	CONFIRM
A039	B-ducts	Return air from rooms via door undercut / door transfer grilles to the return grille near the unit	ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal)	CONFIRM
A040	B-pipes	Refrigerant pipe sizes from GENERIC table (references/pipe_sizing.json) - replace with the selected brand's table	SBC 501 Ch.11 §1107 (refrigerant piping); ASHRAE 15; manufacturer piping tables	CONFIRM
A041	B-pipes	Refrigerant length = route x (1+10%) + vertical rise to roof (+1.5 m allowance)	designer default	default
A042	B-pipes	Condensate to nearest wet-room floor drain / stack with trap; min 25 mm, slope 1.0 %	SBC 501 §307.2.1 (disposal), §307.2.2 Table 307.2.2 (drain size), slope >= 1 %	CONFIRM

2. Room schedule (Gate 1 approved)

ID	Room	Type	Area m²	Ceiling m	Exterior walls (orient. m)	Windows	Roof exp.	Cooled
GF-01	MAJLIS	majlis	27.79	3.0	N 5.7, W 4.68	W1 N 1.8x1.5, W2 W 1.2x1.5	0%	yes
GF-02	ENTRANCE HALL	hall	16.55	3.0	N 3.41	-	0%	yes
GF-03	LIVING	living	30.32	3.0	N 6.21, E 4.7	W4 N 2.4x2.1, W1 E 1.8x1.5	0%	yes
GF-04	CORRIDOR	corridor	7.16	3.0	-	-	0%	yes
GF-05	DINING	dining	21.31	3.0	W 4.35	W1 W 1.8x1.5	0%	yes
GF-06	KITCHEN	kitchen	21.92	3.0	-	-	0%	yes
GF-07	MAID ROOM	maid_room	12.99	3.0	E 3.84	W2 E 1.2x1.5	0%	yes
GF-08	STAIR	stair	12.54	3.0	S 2.91	-	0%	yes
GF-09	MAID WC	bath	6.73	3.0	S 2.34	W3 S 0.6x0.6	0%	no
GF-10	SHAFT	shaft	2.11	3.0	E 2.75, S 0.75	-	0%	no
GF-11	GUEST WC	bath	5.19	3.0	S 2.19, W 2.25	W3 S 0.6x0.6	0%	no
GF-12	STORE	store	5.43	3.0	S 2.38	-	0%	no
GF-13	LAUNDRY	laundry	7.87	3.0	S 4.33	W2 S 1.2x1.5	0%	no
FF-01	MASTER BATH	bath	8.09	3.0	N 2.85	W3 N 0.6x0.6	100%	no
FF-02	BATH 2	bath	6.31	3.0	N 2.25, E 2.69	W3 E 0.6x0.6	100%	no

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FF-03	MASTER BEDROOM	master_bedroom	27.79	3.0	N 5.7, W 4.67	W1 N 1.8x1.5, W2 W 1.2x1.5	100%	yes
FF-04	DRESSING	dressing	5.73	3.0	-	-	100%	yes
FF-05	BEDROOM 2	bedroom	21.37	3.0	N 4.35	W1 N 1.8x1.5	100%	yes
FF-06	LINEN	store	4.35	3.0	E 1.91	-	100%	no
FF-07	LOBBY	corridor	7.23	3.0	-	-	100%	yes
FF-08	BEDROOM 3	bedroom	19.76	3.0	-	-	100%	yes
FF-09	BEDROOM 4	bedroom	13.04	3.0	E 3.85	W1 E 1.8x1.5	100%	yes
FF-10	FAMILY LIVING	living	32.64	3.0	S 4.67, W 6.69	W4 W 2.4x2.1, W2 S 1.2x1.5	100%	yes
FF-11	STAIR	stair	12.46	3.0	S 2.91	-	100%	yes
FF-12	BATH 4	bath	6.73	3.0	S 2.34	W3 S 0.6x0.6	100%	no
FF-13	SHAFT	shaft	2.11	3.0	E 2.75, S 0.75	-	100%	no
FF-14	BATH 3	bath	10.05	3.0	S 4.33	W3 S 0.6x0.6	100%	no

3. Cooling and heating loads

Method: CLTD/SCL/CLF (ASHRAE Fundamentals 1997 Ch.28). Conduction Q = U·A·CLTDc with CLTDc = (CLTD + LM)·K + (25.5 − Ti) + (Tm − 29.4), Tm = To − DR/2. Glass solar Q = A·SC·SHGF·CLF. People/lights/equipment with CLF = 1.0. Infiltration Qs = Cs·V·ΔT, Ql = Cl·V·ΔW. Loads are computed for solar hours 8-20; each room is sized at its own peak, zones and plant at the coincident (block) peak. Heating: Q = ΣU·A·(Ti − To,w) + infiltration + slab edge, no internal gains.

Code reference: ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36; ASHRAE Fundamentals 2021 Ch.16 §'Residential air change method'

ID	Room	m²	Walls	Roof	Glass cond.	Glass solar	People S	Lights	Equip S	Infil S	Sensible W	Latent W	Total W	W/m²	Peak	Supply	Heating W
GF-01	MAJLIS	27.79	273.0	0.0	205.0	370.0	840.0	333.0	111.0	368.0	2501	540	3041	109.4	18:00	199 L/s	828
GF-02	ENTRANCE HALL	16.55	81.0	0.0	0.0	0.0	70.0	132.0	0.0	170.0	453	45	498	30.1	15:00	36 L/s	243
GF-03	LIVING	30.32	246.0	0.0	376.0	272.0	560.0	303.0	243.0	468.0	2468	360	2828	93.3	15:00	196 L/s	980
GF-04	CORRIDOR	7.16	0.0	0.0	0.0	0.0	0.0	43.0	0.0	0.0	43	0	43	6.0	8:00	3 L/s	0
GF-05	DINING	21.31	123.0	0.0	123.0	418.0	630.0	213.0	64.0	188.0	1759	405	2164	101.5	18:00	140 L/s	400
GF-06	KITCHEN	21.92	0.0	0.0	0.0	0.0	160.0	219.0	1500.0	0.0	1879	660	2539	115.8	8:00	149 L/s	0
GF-07	MAID ROOM	12.99	81.0	0.0	55.0	286.0	60.0	104.0	65.0	70.0	720	25	745	57.4	9:00	57 L/s	294
GF-08	STAIR	12.54	69.0	0.0	0.0	0.0	0.0	75.0	0.0	126.0	270	0	270	21.6	16:00	22 L/s	197
FF-03	MASTER BEDROOM	27.79	273.0	172.0	205.0	370.0	120.0	222.0	167.0	368.0	1897	50	1947	70.1	18:00	151 L/s	728
FF-04	DRESSING	5.73	0.0	35.0	0.0	0.0	0.0	46.0	0.0	0.0	81	0	81	14.2	17:00	6 L/s	17
FF-05	BEDROOM 2	21.37	85.0	123.0	131.0	83.0	120.0	171.0	107.0	220.0	1039	50	1089	51.0	15:00	83 L/s	387
FF-07	LOBBY	7.23	0.0	45.0	0.0	0.0	0.0	43.0	0.0	0.0	88	0	88	12.2	17:00	7 L/s	21
FF-08	BEDROOM 3	19.76	0.0	122.0	0.0	0.0	120.0	158.0	99.0	0.0	499	50	549	27.8	17:00	40 L/s	57
FF-09	BEDROOM 4	13.04	75.0	36.0	83.0	428.0	120.0	104.0	65.0	70.0	982	50	1032	79.1	9:00	78 L/s	295
FF-10	FAMILY LIVING	32.64	285.0	202.0	312.0	821.0	630.0	326.0	261.0	432.0	3270	405	3675	112.6	18:00	260 L/s	889
FF-11	STAIR	12.46	69.0	76.0	0.0	0.0	0.0	75.0	0.0	126.0	345	0	345	27.7	16:00	27 L/s	182

Conditioned area 290.6 m²; sum of room peaks 20.9 kW; block peak 20.5 kW at 17:00 (70 W/m²); heating 5.5 kW. Safety factors are applied at equipment selection (section 5).

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3.1 Worked example - FF-10 FAMILY LIVING at 18:00

Item	Calculation
Wall S	U 0.4 × A 15.01 m² (gross 16.81)
Wall W	U 0.4 × A 19.04 m² (gross 24.08)
Window W4 W	cond. U 2.67 × 5.04 m²; solar 5.04 × SC 0.287 × SHGF 690 × CLF
Window W2 S	cond. U 2.67 × 1.8 m²; solar 1.8 × SC 0.287 × SHGF 110 × CLF
Roof	U 0.2 × 32.64 m²
People	9 persons
Infiltration	0.75 ACH × 97.9 m³ = 20.4 L/s
wall	285.0 W
roof	202.0 W
glass_cond	312.0 W
glass_solar	821.0 W
people_s	630.0 W
lights	326.0 W
equip_s	261.0 W
infil_s	432.0 W
people_l	405 W
equip_l	0 W
infil_l	0 W
Room sensible / latent / total	3270 / 405 / 3675 W
Supply airflow	3270 W / (1.143 × (24.0 − 13.0)) = 260.0 L/s

4. Ventilation and exhaust

Ventilation rate procedure: Vbz = Rp·Pz + Ra·Az; Voz = Vbz / Ez; single-zone Vot = Voz; multiple-zone Vot = (D·ΣRp·Pz + ΣRa·Az)/Ev.

Code reference: SBC 501 §403.3.1.1 & Table 403.3.1.1; ASHRAE 62.1-2019 §6.2.2.1 Eq.6-1, Table 6-1; ASHRAE 62.1-2019 §6.2.2.2 Table 6-2 (Ez); ASHRAE 62.1-2019 §6.2.4.3.1 simplified procedure (Ev)

Zone	Room	Category	Pz	Az m²	Rp·Pz	Ra·Az	Vbz L/s	Ez	Ev	Zone Vot L/s	OA method
Z-GF-01	GF-06	dwelling	2	21.92	5.0	6.6	11.6	0.8	1.0	14.5	transfer air from Z-GF-04 (OA added to that zone)
Z-GF-02	GF-07	dwelling	1	12.99	2.5	3.9	6.4	0.8	1.0	8.0	through-wall fresh air fan
Z-GF-03	GF-01	dwelling	12	27.79	30.0	8.3	38.3	0.8	1.0	47.9	ducted to unit return plenum
Z-GF-04	GF-05	dwelling	9	21.31	22.5	6.4	28.9	0.8	0.75	106.3	ducted to unit return plenum
Z-GF-04	GF-08	corridor	0	12.54	0.0	3.8	3.8	0.8	0.75	106.3	ducted to unit return plenum
Z-GF-04	GF-04	corridor	0	7.16	0.0	2.1	2.1	0.8	0.75	106.3	ducted to unit return plenum
Z-GF-04	GF-02	corridor	1	16.55	0.0	5.0	5.0	0.8	0.75	106.3	ducted to unit return plenum
Z-GF-04	GF-03	dwelling	8	30.32	20.0	9.1	29.1	0.8	0.75	106.3	ducted to unit return plenum
Z-FF-01	FF-03	dwelling	2	27.79	5.0	8.3	13.3	0.8	0.75	20.1	ducted to unit return plenum
Z-FF-01	FF-04	dwelling	0	5.73	0.0	1.7	1.7	0.8	0.75	20.1	ducted to unit return plenum
Z-FF-02	FF-10	dwelling	9	32.64	22.5	9.8	32.3	0.8	0.75	92.6	ducted to unit

											return plenum
Z-FF-02	FF-11	corridor	0	12.46	0.0	3.7	3.7	0.8	0.75	92.6	ducted to unit return plenum
Z-FF-02	FF-07	corridor	0	7.23	0.0	2.2	2.2	0.8	0.75	92.6	ducted to unit return plenum
Z-FF-02	FF-08	dwelling	2	19.76	5.0	5.9	10.9	0.8	0.75	92.6	ducted to unit return plenum
Z-FF-02	FF-05	dwelling	2	21.37	5.0	6.4	11.4	0.8	0.75	92.6	ducted to unit return plenum
Z-FF-02	FF-09	dwelling	2	13.04	5.0	3.9	8.9	0.8	0.75	92.6	ducted to unit return plenum

Room	Name	Rule	Exhaust L/s	Mode	Clause
GF-06	KITCHEN	kitchen_residential	50.0	intermittent	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
GF-09	MAID WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-11	GUEST WC	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
GF-13	LAUNDRY	laundry	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-01	MASTER BATH	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-02	BATH 2	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-12	BATH 4	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
FF-14	BATH 3	toilet_private	25.0	intermittent	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)

Building outdoor air 275 L/s; exhaust 225 L/s (net +50 L/s). ASHRAE 62.2 whole-house cross-check: 65 L/s - design OA exceeds it.

5. Zoning and equipment selection

Required capacity = coincident zone load (rooms + outdoor air) × (1 + SF), SF cooling 10%, heating 10%. Nominal capacities × 0.88 for high-ambient (T3) operation. Selection = smallest catalogue unit meeting total, sensible and airflow.

Code reference: Designer allowance - not a code requirement; SASO 2663 / SBC 601 Table C403.3.2 minimum efficiency

Zone	Rooms	System	Peak h	Rooms kW	OA kW	Block kW	SF	Req. kW	Req. sens kW	Req. air	Selected	Derated kW	Sens kW	Unit air	OA %	Input kW	Heat kW sel/req
Z-GF-01	GF-06	wall_split	15	2.54	0.36	2.9	10%	3.19	2.46	149 L/s	IU-GF-01 WS-18	4.66	3.45	250 L/s	6%	1.61	5.6 / 0.26
Z-GF-02	GF-07	wall_split	15	0.67	0.2	0.86	10%	0.95	0.92	57 L/s	IU-GF-02 WS-09	2.29	1.74	140 L/s	6%	0.76	2.9 / 0.47
Z-GF-03	GF-01	ducted_s	16	2.98	1.16	4.14	10%	4.55	3.96	199 L/s	IU-GF-03	6.16	4.62	380 L/s	13%	2.19	7.6 / 1.78

		plit									DS-24						
Z-GF-04	GF-05, GF-08, GF-04, GF-02, GF-03	ducted_split	16	5.67	2.58	8.25	10%	9.07	8.18	432 L/s	IU-GF-04 DS-48	12.32	9.12	750 L/s	14%	4.52	15.0 / 3.93
Z-FF-01	FF-03, FF-04	ducted_split	17	2.01	0.46	2.47	10%	2.72	2.66	181 L/s	IU-FF-01 DS-24	6.16	4.62	380 L/s	5%	2.19	7.6 / 1.18
Z-FF-02	FF-10, FF-11, FF-07, FF-08, FF-05, FF-09	ducted_split	16	6.47	2.24	8.72	10%	9.59	8.98	520 L/s	IU-FF-02 DS-48	12.32	9.12	750 L/s	12%	4.52	15.0 / 3.69

Tag	Model	System	Serves	Nominal	Input kW	Location
ODU-01	WS-O-18	wall_split	IU-GF-01	5.3 kW	1.61	RF
ODU-02	WS-O-09	wall_split	IU-GF-02	2.6 kW	0.76	RF
ODU-03	DS-O-24	ducted_split	IU-GF-03	7.0 kW	2.19	RF
ODU-04	DS-O-48	ducted_split	IU-GF-04	14.0 kW	4.52	RF
ODU-05	DS-O-24	ducted_split	IU-FF-01	7.0 kW	2.19	RF
ODU-06	DS-O-48	ducted_split	IU-FF-02	14.0 kW	4.52	RF

6. Duct sizing and air terminals

Equal friction method, target 0.80 Pa/m, velocity limits (m/s) {'main': 5.0, 'branch': 3.5, 'runout': 2.5, 'return': 3.5, 'fresh_air': 4.0, 'exhaust': 4.0}. Darcy-Weisbach with Altshul-Tsal friction factor, roughness 0.09 mm; rectangular sizes from De = 1.30(ab)^0.625/(a+b)^0.25; fittings +50% equivalent length.

Code reference: ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal); SBC 501 §603.4 metallic ducts; SMACNA 2005 Tables 2-1/2-2 (gauge)

Z-GF-03 - IU-GF-03 (GF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-GF-001	main	-	380 L/s	300x300	328	4.22 m/s	0.72 Pa/m	4.72	5.1 Pa	26 ga (0.55 mm)
D-GF-002	runout	GF-01	190 L/s	Ø350 flex	350	1.97 m/s	0.15 Pa/m	1.47	0.3 Pa	flex
D-GF-003	runout	GF-01	190 L/s	Ø350 flex	350	1.97 m/s	0.15 Pa/m	1.47	0.3 Pa	flex
D-GF-004	main	-	380 L/s	400x300	378	3.17 m/s	0.36 Pa/m	1.2	0.6 Pa	24 ga (0.70 mm)
D-GF-005	main	-	48 L/s	Ø150 round	150	2.71 m/s	0.76 Pa/m	2.4	2.7 Pa	26 ga (0.55 mm)

Critical path: required ESP 71 Pa vs available 75 Pa - OK.

Z-GF-04 - IU-GF-04 (GF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-GF-006	main	-	403 L/s	350x300	354	3.84 m/s	0.55 Pa/m	3.63	3.0 Pa	24 ga (0.70 mm)
D-GF-007	main	-	340 L/s	300x300	328	3.78 m/s	0.59 Pa/m	0.1	0.1 Pa	26 ga (0.55 mm)
D-GF-008	main	-	295 L/s	300x250	299	3.93 m/s	0.71 Pa/m	1.0	1.1 Pa	26 ga (0.55 mm)
D-GF-009	main	-	52 L/s	150x150	164	2.32 m/s	0.58 Pa/m	2.38	2.1 Pa	26 ga (0.55 mm)
D-GF-010	branch	GF-05	243 L/s	300x300	328	2.70 m/s	0.32 Pa/m	3.97	1.9 Pa	26 ga (0.55 mm)
D-GF-011	runout	GF-05	121 L/s	Ø250 flex	250	2.47 m/s	0.34 Pa/m	1.22	0.6 Pa	flex
D-GF-012	runout	GF-05	121 L/s	Ø250 flex	250	2.47 m/s	0.34 Pa/m	1.22	0.6 Pa	flex
D-GF-013	branch	GF-02	62 L/s	200x150	189	2.08 m/s	0.40 Pa/m	1.25	0.8 Pa	26 ga (0.55 mm)
D-GF-014	runout	GF-02	31 L/s	Ø150 flex	150	1.77 m/s	0.35 Pa/m	1.22	0.6 Pa	flex
D-GF-015	runout	GF-02	31 L/s	Ø150 flex	150	1.77 m/s	0.35 Pa/m	1.22	0.6 Pa	flex
D-GF-016	branch	GF-03	340 L/s	400x300	378	2.84 m/s	0.30 Pa/m	6.22	2.8 Pa	24 ga (0.70 mm)

D-GF-017	runout	GF-03	170 L/s	Ø300 round	300	2.41 m/s	0.26 Pa/m	1.59	0.6 Pa	26 ga (0.55 mm)
D-GF-018	runout	GF-03	170 L/s	Ø300 round	300	2.41 m/s	0.26 Pa/m	1.59	0.6 Pa	26 ga (0.55 mm)
D-GF-019	main	-	750 L/s	850x300	534	2.94 m/s	0.23 Pa/m	1.2	0.4 Pa	22 ga (0.85 mm)
D-GF-020	main	-	106 L/s	Ø200 round	200	3.38 m/s	0.80 Pa/m	6.09	7.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 72 Pa vs available 100 Pa - OK.

Z-FF-01 - IU-FF-01 (FF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-FF-001	main	-	317 L/s	300x300	328	3.52 m/s	0.52 Pa/m	4.43	3.4 Pa	26 ga (0.55 mm)
D-FF-002	branch	FF-03	317 L/s	350x300	354	3.02 m/s	0.36 Pa/m	1.43	0.8 Pa	24 ga (0.70 mm)
D-FF-003	runout	FF-03	158 L/s	Ø300 flex	300	2.24 m/s	0.23 Pa/m	1.47	0.5 Pa	flex
D-FF-004	runout	FF-03	158 L/s	Ø300 flex	300	2.24 m/s	0.23 Pa/m	1.47	0.5 Pa	flex
D-FF-005	main	-	380 L/s	400x300	378	3.17 m/s	0.36 Pa/m	1.2	0.6 Pa	24 ga (0.70 mm)
D-FF-006	main	-	20 L/s	Ø125 round	125	1.64 m/s	0.39 Pa/m	3.89	2.3 Pa	26 ga (0.55 mm)

Critical path: required ESP 70 Pa vs available 75 Pa - OK.

Z-FF-02 - IU-FF-02 (FF)

Segment	Type	Serves	Flow	Size	De mm	Velocity	Friction	Length m	Δp	Gauge
D-FF-007	riser	-	750 L/s	600x300	457	4.17 m/s	0.49 Pa/m	0.76	0.6 Pa	24 ga (0.70 mm)
D-FF-008	main	-	375 L/s	300x300	328	4.16 m/s	0.70 Pa/m	4.34	4.6 Pa	26 ga (0.55 mm)
D-FF-009	main	-	289 L/s	300x250	299	3.85 m/s	0.69 Pa/m	3.75	3.9 Pa	26 ga (0.55 mm)
D-FF-010	main	-	232 L/s	250x250	273	3.71 m/s	0.71 Pa/m	1.05	1.1 Pa	26 ga (0.55 mm)
D-FF-011	main	-	113 L/s	200x200	219	2.81 m/s	0.57 Pa/m	2.92	2.5 Pa	26 ga (0.55 mm)
D-FF-012	branch	FF-10	375 L/s	400x300	378	3.12 m/s	0.35 Pa/m	0.9	0.5 Pa	24 ga (0.70 mm)
D-FF-013	runout	FF-10	125 L/s	Ø300 round	300	1.77 m/s	0.15 Pa/m	1.97	0.4 Pa	26 ga (0.55 mm)
D-FF-014	runout	FF-10	125 L/s	Ø300 round	300	1.77 m/s	0.15 Pa/m	2.8	0.6 Pa	26 ga (0.55 mm)
D-FF-015	runout	FF-10	125 L/s	Ø300 round	300	1.77 m/s	0.15 Pa/m	3.12	0.7 Pa	26 ga (0.55 mm)
D-FF-016	branch	FF-11	43 L/s	150x150	164	1.92 m/s	0.41 Pa/m	2.68	1.7 Pa	26 ga (0.55 mm)
D-FF-017	branch	FF-07	43 L/s	150x150	164	1.92 m/s	0.41 Pa/m	0.75	0.5 Pa	26 ga (0.55 mm)
D-FF-018	branch	FF-08	57 L/s	150x150	164	2.54 m/s	0.68 Pa/m	0.25	0.3 Pa	26 ga (0.55 mm)
D-FF-019	branch	FF-05	119 L/s	200x200	219	2.97 m/s	0.63 Pa/m	4.47	4.2 Pa	26 ga (0.55 mm)
D-FF-020	runout	FF-05	60 L/s	Ø200 flex	200	1.89 m/s	0.28 Pa/m	1.22	0.5 Pa	flex
D-FF-021	runout	FF-05	60 L/s	Ø200 flex	200	1.89 m/s	0.28 Pa/m	1.22	0.5 Pa	flex
D-FF-022	main	-	750 L/s	850x300	534	2.94 m/s	0.23 Pa/m	1.2	0.4 Pa	22 ga (0.85 mm)
D-FF-023	main	-	93 L/s	Ø200 round	200	2.95 m/s	0.62 Pa/m	6.12	5.7 Pa	26 ga (0.55 mm)

Critical path: required ESP 73 Pa vs available 100 Pa - OK.

6.1 Supply diffuser schedule

Tag	Room	Model	Flow	NC	Throw m
SD-GF-01	GF-01	SD-250	190 L/s	28	2.9
SD-GF-02	GF-01	SD-250	190 L/s	28	2.9
SD-GF-03	GF-05	SD-200	121 L/s	26	2.2
SD-GF-04	GF-05	SD-200	121 L/s	26	2.2
SD-GF-05	GF-08	SD-150	52 L/s	18	1.3
SD-GF-06	GF-04	SD-150	52 L/s	18	1.3
SD-GF-07	GF-02	SD-150	31 L/s	11	0.8
SD-GF-08	GF-02	SD-150	31 L/s	11	0.8
SD-GF-09	GF-03	SD-250	170 L/s	26	2.6
SD-GF-10	GF-03	SD-250	170 L/s	26	2.6
SD-FF-01	FF-03	SD-250	158 L/s	24	2.4

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SD-FF-02	FF-03	SD-250	158 L/s	24	2.4
SD-FF-03	FF-04	SD-150	63 L/s	22	1.5
SD-FF-04	FF-10	SD-200	125 L/s	27	2.3
SD-FF-05	FF-10	SD-200	125 L/s	27	2.3
SD-FF-06	FF-10	SD-200	125 L/s	27	2.3
SD-FF-07	FF-11	SD-150	43 L/s	15	1.0
SD-FF-08	FF-07	SD-150	43 L/s	15	1.0
SD-FF-09	FF-08	SD-150	57 L/s	20	1.4
SD-FF-10	FF-05	SD-150	60 L/s	21	1.4
SD-FF-11	FF-05	SD-150	60 L/s	21	1.4
SD-FF-12	FF-09	SD-200	113 L/s	24	2.1

Code reference: ASHRAE Applications 2019 Ch.49 Table 1 (design NC)

6.2 Exhaust systems

Fan	Room	Flow	Model	Duct	Length m	Velocity	ESP req.	Discharge	Clause
EF-GF-01	GF-06	50 L/s	EF-50	Ø150	4.41	2.83 m/s	46 Pa	wall_louver	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
EF-GF-02	GF-09	25 L/s	EF-25	Ø125	1.48	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-03	GF-11	25 L/s	EF-25	Ø125	1.12	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-GF-04	GF-13	25 L/s	EF-25	Ø125	1.78	2.04 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-01	FF-01	25 L/s	EF-25	Ø125	2.27	2.04 m/s	42 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-02	FF-02	25 L/s	EF-25	Ø125	1.22	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-03	FF-12	25 L/s	EF-25	Ø125	1.48	2.04 m/s	41 Pa	wall_louver	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
EF-FF-04	FF-14	25 L/s	EF-25	Ø125	1.24	2.04 m/s	41 Pa	wall_louver	SBC 501 Table

									403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)

7. Piping

7.1 Refrigerant piping

Run	Unit → ODU	Capacity	Liquid / gas OD mm	Horiz. floor m	Vertical m	Roof m	Total m	Limit m	Status
RP-IU-GF-01	IU-GF-01 → ODU-01	5.3 kW	6.35 / 12.7 (1/4" / 1/2")	10.39	8.7	0.65	21.7	25	OK
RP-IU-GF-02	IU-GF-02 → ODU-02	2.6 kW	6.35 / 9.52 (1/4" / 3/8")	2.63	8.7	1.9	14.6	25	OK
RP-IU-GF-03	IU-GF-03 → ODU-03	7.0 kW	9.52 / 15.88 (3/8" / 5/8")	15.51	8.7	2.2	29.1	50	OK
RP-IU-GF-04	IU-GF-04 → ODU-04	14.0 kW	9.52 / 15.88 (3/8" / 5/8")	13.09	8.7	3.45	27.8	50	OK
RP-IU-FF-01	IU-FF-01 → ODU-05	7.0 kW	9.52 / 15.88 (3/8" / 5/8")	14.39	5.1	3.5	25.3	50	OK
RP-IU-FF-02	IU-FF-02 → ODU-06	14.0 kW	9.52 / 15.88 (3/8" / 5/8")	13.09	5.1	3.75	24.1	50	OK

Code reference: SBC 501 Ch.11 §1107 (refrigerant piping); ASHRAE 15; manufacturer piping tables

7.3 Condensate drains

Run	Unit	To	Code min mm	Design mm	Length m	Slope %
CD-IU-GF-01	IU-GF-01	GF-06 KITCHEN (floor drain / waste stack with trap)	20	25	2.54	1.0
CD-IU-GF-02	IU-GF-02	GF-10 SHAFT (floor drain / waste stack with trap)	20	25	2.9	1.0
CD-IU-GF-03	IU-GF-03	GF-06 KITCHEN (floor drain / waste stack with trap)	20	25	8.14	1.0
CD-IU-GF-04	IU-GF-04	GF-06 KITCHEN (floor drain / waste stack with trap)	20	25	5.48	1.0
CD-IU-FF-01	IU-FF-01	FF-01 MASTER BATH (floor drain / waste stack with trap)	20	25	2.69	1.0
CD-IU-FF-02	IU-FF-02	FF-01 MASTER BATH (floor drain / waste stack with trap)	20	25	6.24	1.0

Code reference: SBC 501 §307.2.1 (disposal), §307.2.2 Table 307.2.2 (drain size), slope >= 1 %

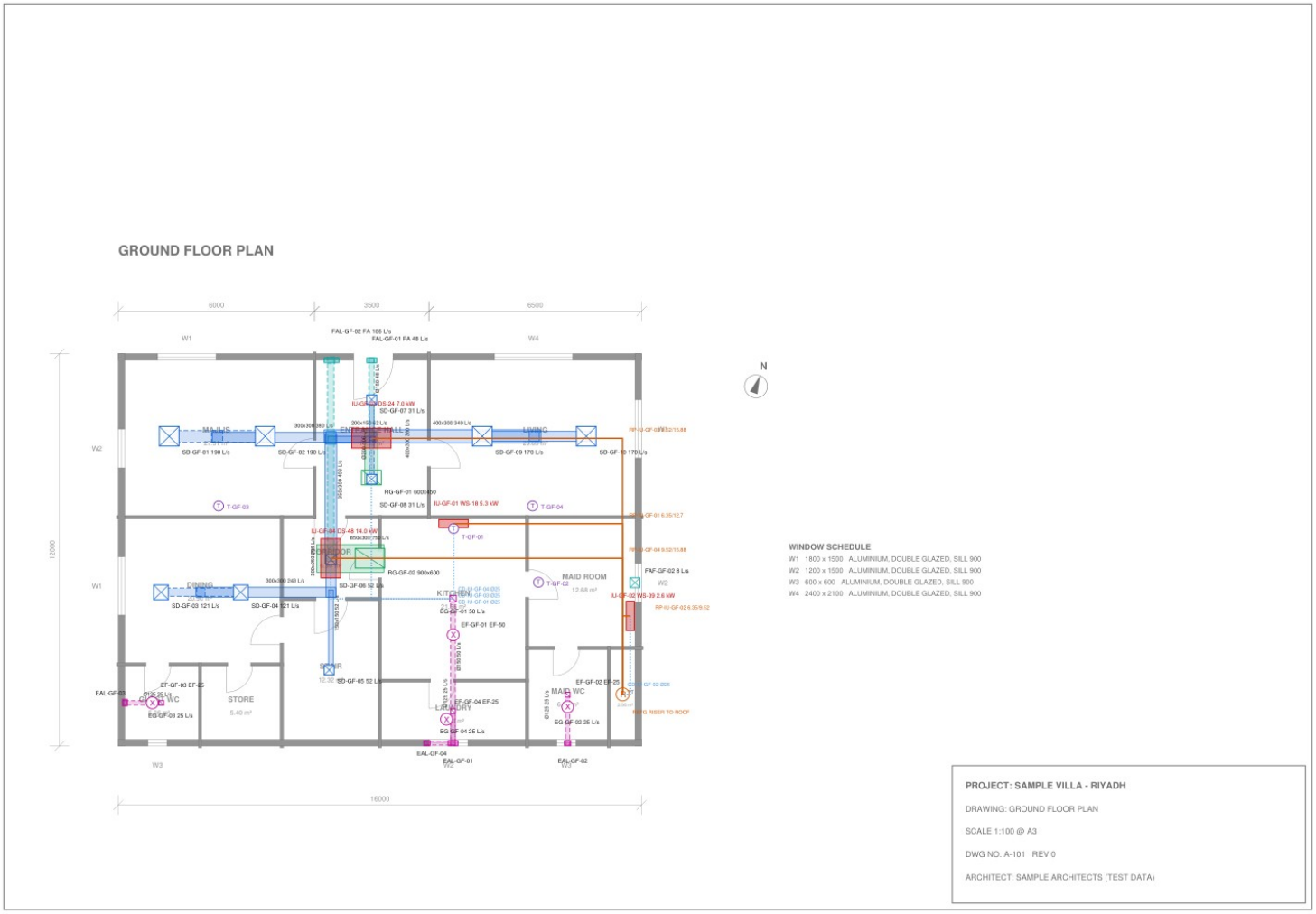
8. Design checks

Check	Status	Detail
Every room served	PASS	16 conditioned rooms served, 8 exhaust rooms; not served by design (stores/shafts): GF-10, GF-12, FF-06, FF-13
No duct crossing shaft/stair/column unresolved	PASS	trunk routes chosen clear of shafts, stairs (not served) and columns; beams not on plan - coordinate
Airflows balance	PASS	supply = unit airflow in all ducted zones; building OA 274.9 L/s vs exhaust 225.0 L/s (net +50 L/s)
All items tagged (unique)	PASS	73 tags
Every drawing item in BOQ	PASS	85 drawn tags all in BOQ (pipe runs 6 measured by size in section G)
Unit external static pressure	PASS	all ducted units have ESP margin
Refrigerant length / height limits	PASS	within generic limits (verify with brand)
Diffuser noise (NC)	PASS	all diffusers at or below room NC target
Fresh-air fraction	PASS	OA fraction within limit for untreated fresh air
Open questions	PASS	none

9. Code clause register

Rule	Clause applied
design_conditions	SBC 602 §302 / SBC 601 §302.1 (design conditions); ASHRAE Fundamentals 2021 Ch.14 (0.4 % cooling, 99.6 % heating)
envelope_u	SBC 602 Ch.4 prescriptive envelope table / SBC 601 Ch.5 (max U-factor, SHGC)
load_method	ASHRAE Fundamentals 1997 Ch.28 CLTD/SCL/CLF method, Tables 30-36
infiltration	ASHRAE Fundamentals 2021 Ch.16 §'Residential air change method'
internal_gains	ASHRAE Fundamentals 2021 Ch.18 Table 1 (people), Tables 2-3 (lighting), Tables 5-9 (equipment)
oa_rate	SBC 501 §403.3.1.1 & Table 403.3.1.1; ASHRAE 62.1-2019 §6.2.2.1 Eq.6-1, Table 6-1
oa_zone_eff	ASHRAE 62.1-2019 §6.2.2.2 Table 6-2 (Ez)
oa_system_eff	ASHRAE 62.1-2019 §6.2.4.3.1 simplified procedure (Ev)
exhaust_toilet	SBC 501 Table 403.3.1.1 (toilet rooms, private dwellings: 25 L/s intermittent or 10 L/s continuous)
exhaust_kitchen	SBC 501 Table 403.3.1.1 (kitchens, private dwellings: 50 L/s intermittent or 12 L/s continuous)
exhaust_toilet_public	ASHRAE 62.1-2019 Table 6-4 (toilet public: 25 L/s per WC/urinal)
duct_sizing	ASHRAE Fundamentals 2021 Ch.21 (equal friction method, Eq.18-19 Altshul-Tsal)
duct_construction	SBC 501 §603.4 metallic ducts; SMACNA 2005 Tables 2-1/2-2 (gauge)
duct_insulation	SBC 601 §C403.11.1 / ASHRAE 90.1-2019 §6.4.4.1.2, Table 6.8.2
pipe_insulation	SBC 601 §C403.11.3 / ASHRAE 90.1-2019 §6.4.4.1.3, Table 6.8.3-1/-2
refrigerant_piping	SBC 501 Ch.11 §1107 (refrigerant piping); ASHRAE 15; manufacturer piping tables
chw_piping	ASHRAE Fundamentals 2021 Ch.22 (water velocity / friction limits)
condensate	SBC 501 §307.2.1 (disposal), §307.2.2 Table 307.2.2 (drain size), slope >= 1 %
equipment_efficiency	SASO 2663 / SBC 601 Table C403.3.2 minimum efficiency
noise	ASHRAE Applications 2019 Ch.49 Table 1 (design NC)
safety_factor	Designer allowance - not a code requirement
controls	SBC 601 §C403.4 / ASHRAE 90.1 §6.4.3 (thermostatic controls, OA damper shut-off)

Appendix - sample sheet (M-101)



PRELIMINARY DESIGN - FOR COORDINATION ONLY. To be checked and stamped by a qualified HVAC engineer before construction.

HVAC LAYOUT - GROUND FLOOR

N

North 20° from page-up
Scale 1:100 (as architectural sheet)

LEGEND

- Supply duct (size, flow) / flex dashed
- Supply diffuser SD
- Return duct
- Return grille RG
- Fresh air duct / louver FAL
- Exhaust duct / grille EG / fan EF
- Refrigerant pipe (liquid/gas mm)
- Condensate drain
- Chilled water pipe
- Indoor / outdoor unit
- Thermostat / controller

EQUIPMENT ON THIS SHEET

IU-GF-01 WS-18 5.3 kW 250 L/s Z-GF-01
IU-GF-02 WS-09 2.6 kW 140 L/s Z-GF-02
IU-GF-03 DS-24 7.0 kW 380 L/s Z-GF-03
IU-GF-04 DS-48 14.0 kW 750 L/s Z-GF-04
EF-GF-01 EF-50 50 L/s GF-06
EF-GF-02 EF-25 25 L/s GF-09
EF-GF-03 EF-25 25 L/s GF-11
EF-GF-04 EF-25 25 L/s GF-13

NOTES

- Duct sizes W x H mm (clear internal); Ø = round.
- Ducts at high level in ceiling void; coordinate with beams.
- Door undercut 20 mm / transfer grilles for return air.
- Refrigerant/condensate routes are indicative.

PROJECT: Sample Villa - Riyadh
LOCATION: Riyadh, Saudi Arabia
SHEET: HVAC LAYOUT - Ground Floor
SHEET NO. M-101 REV P0 2026-10-15
SCALE: 1:100 CODES: Saudi Building Code + ASHRAE
PREPARED WITH hvac-designer skill (automated)

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